

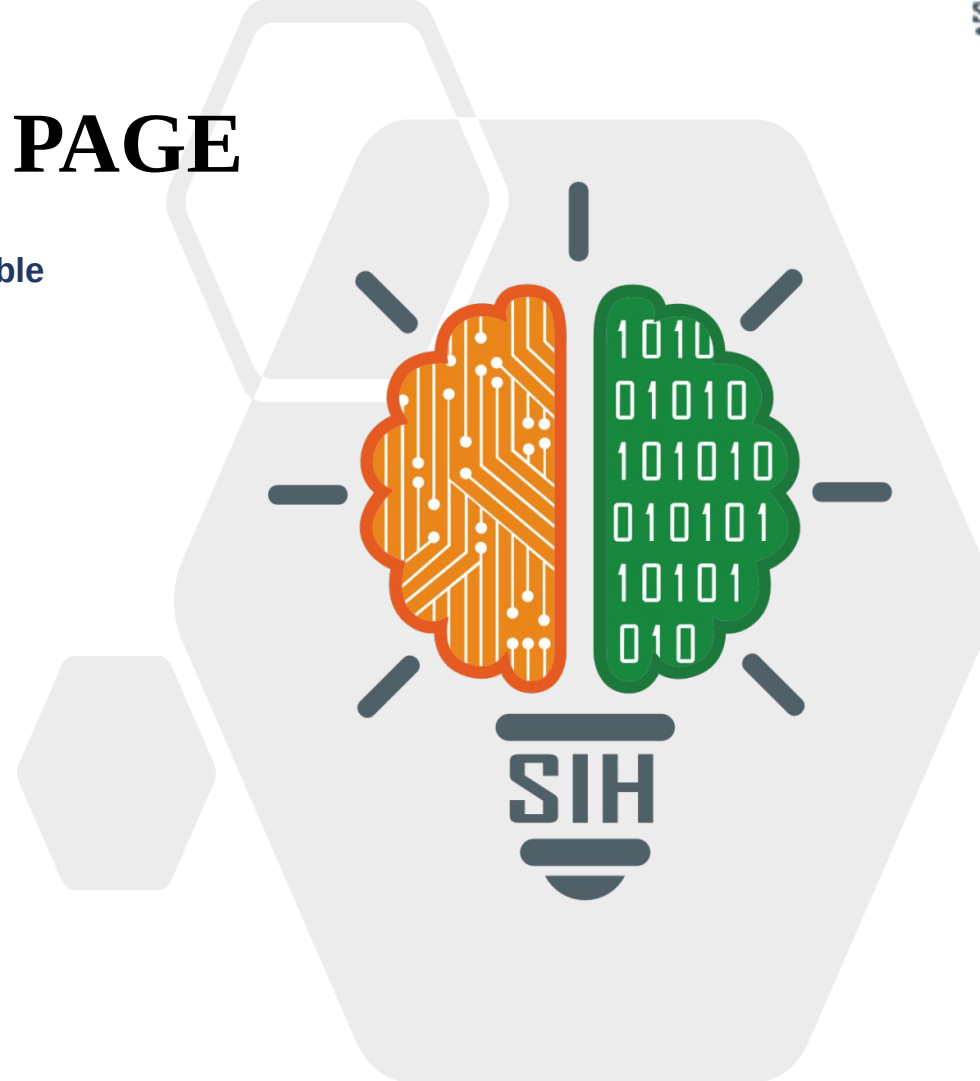
SMART INDIA HACKATHON 2025



TITLE PAGE

Idea: YatraDrishti – Hyper-Personalized AI Ecosystem for Sustainable Tourism & Smart Hospitality

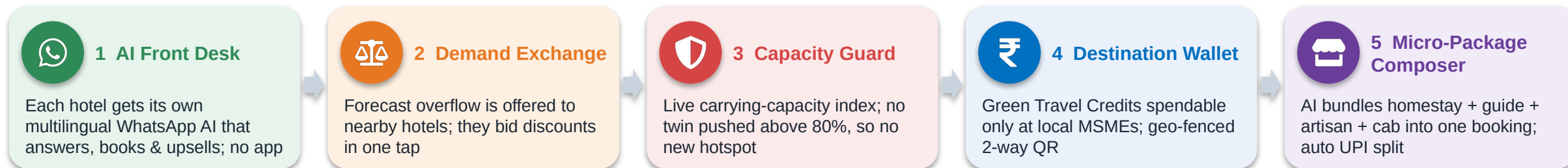
- **Problem Statement ID** – _____
- **Problem Statement Title**- Student Innovation
- **Theme**- Travel & Tourism
- **PS Category**- Software
- **Team ID**- _____
- **Team Name (Registered on portal)** _____



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution

YatraDrishti turns every small hotel's WhatsApp into an **AI front desk** and links them in a live **Tourism Demand Exchange**: before a hotspot overflows, nearby hotels bid for the surplus tourists within safe capacity, and credits keep the money in the local economy.



❖ How it addresses the problem

- **Overcrowded hotspots** → overflow auctioned to nearby hotels in advance
- **15–30% OTA commissions** → direct bookings via the hotel's own AI WhatsApp
- **No live crowd data** → 1-tap WhatsApp pulses from hotels & vendors
- **Invisible artisans & homestays** → bundled into AI micro-packages
- **Tourist money leaks out** → credits recirculate inside the destination

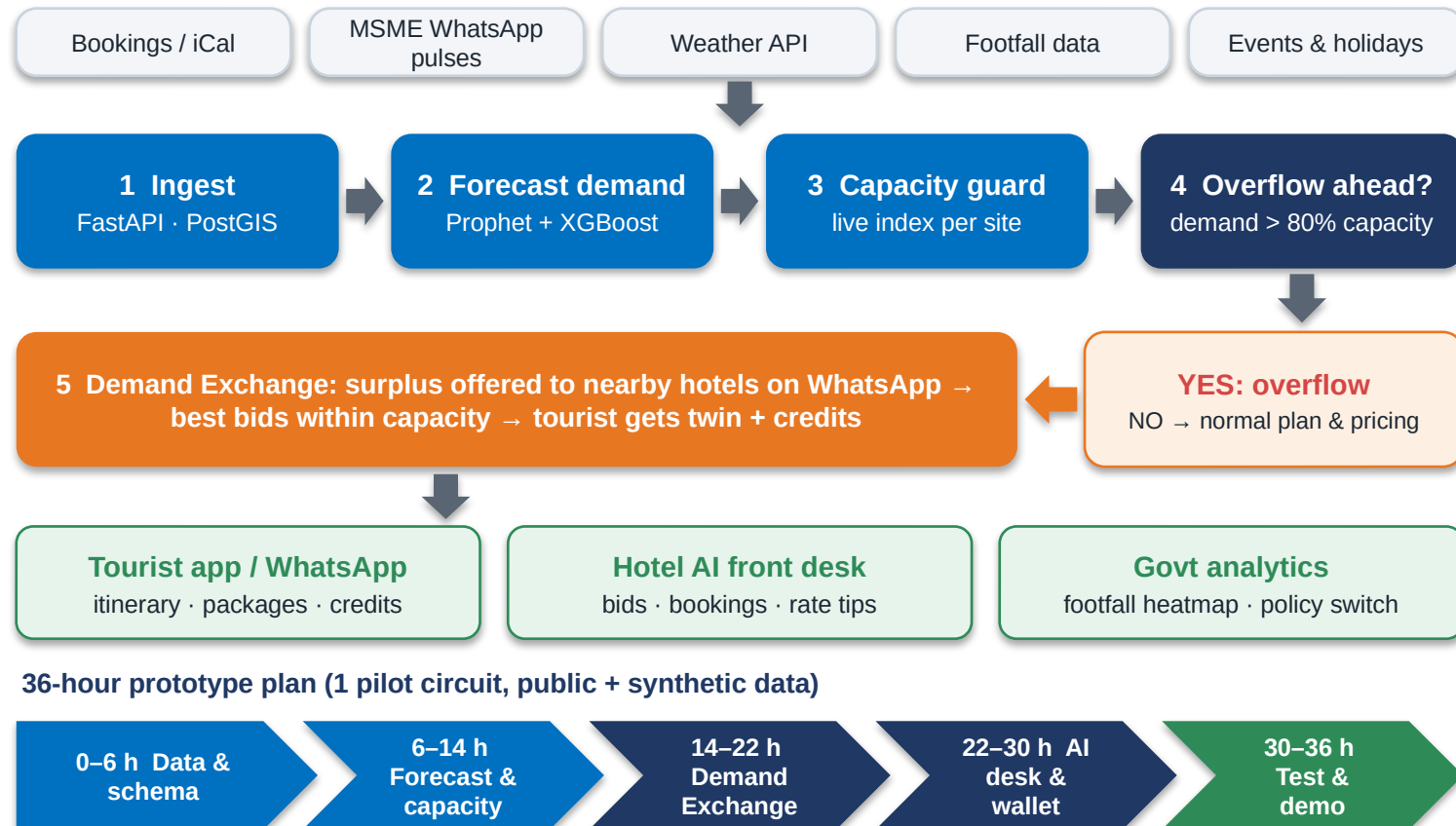
❖ Innovation and uniqueness of the solution

- **Self-funding crowd balancing**: hotels bid for diverted tourists
- **Hotel-in-a-WhatsApp**: multi-tenant AI front desk, already prototyped
- **Human sensor network**: MSME pulses replace tourist tracking
- **Closed-loop local money**: credits spendable only in the destination
- **No overflow cascade**: capacity guard keeps every twin under 80%

Technologies to be used (e.g. programming languages, frameworks, hardware)

Layer	Technology
Frontend	React Native / Flutter · Next.js dashboard
Backend	FastAPI (Python) · Node.js (TypeScript) gateway
WhatsApp	Multi-tenant Baileys AI gateway (already built) → per-hotel AI desk
AI / ML	XGBoost + Prophet · LangChain + Llama-3 / Gemini
Matching	OR-Tools: bid matching under capacity limits
Data	PostgreSQL + PostGIS · Redis · H3 · Docker / K8s
Payments	UPI split settlement (Razorpay Route, test mode)
Tourist UX	AR (ARCore / ARKit) · TensorFlow Lite offline guide
Hardware	None: runs on any smartphone

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Emergency policy switch: one admin flag freezes discounts and reroutes tourists during flood / weather alerts (can be shown live to judges).

Analysis of the feasibility of the idea

Technical ✓

- WhatsApp AI gateway already works
- Per-hotel AI desk = new tenant config, not new code

Data & APIs ✓

- data.tourism.gov.in, OSM, weather, Kaggle hotel data
- Backup: synthetic footfall + 50 hotels

Economic ✓

- Hotel SaaS ≈ ₹1,250–4,200 / month
- Small fee on Exchange wins & packages

Operational ✓

- Nothing to install: hotels use WhatsApp
- Fits existing PMS via iCal

Phase 1: Pilot: Golden Triangle, Puducherry, Himachal

Phase 2: State Tourism Dev. Corporations

Phase 3: Hotel associations & homestay networks

National: Incredible India integration

Potential challenges and risks

Strategies for overcoming these challenges

WhatsApp session keys exposed in code

Revoke linked device, purge .baileys_auth from git, secrets in environment variables

Sessions lost on every restart

Database-backed auth store (PostgreSQL / Redis) with caching & safe key serialisation

WhatsApp rate limits / number ban

Queue, rate limiter, randomised delays; official WhatsApp Business API in production

Cross-tenant data leakage

Per-tenant session factory + JWT tenant binding on every endpoint

Hotels game the Demand Exchange

Bids checked against live inventory, discount caps, hotel reputation score

Fake credit redemptions

Two-way geo-fenced QR: tourist and vendor must be within 50 m

No live crowd data

MSME WhatsApp pulses + anonymised H3 counts; synthetic data as backup

❖ Potential impact on the target audience



Tourists

20–35%

savings on off-peak routes*

- Less crowded trips
- Credits for off-peak choices
- One-booking local packages



Small hotels

+22–30%

low-season yield*

- Own AI desk, no OTA cut
- Win overflow guests by bidding
- AI rate tips on WhatsApp



Artisans & locals

+35%

direct revenue*

- Sold inside micro-packages
- Instant UPI split payouts
- Credits spent at their shops



Governments

up to 40%

less peak congestion*

- Overflow handled in advance
- Live MSME crowd pulses
- Emergency policy switch

❖ Benefits of the solution (social, economic, environmental, etc.)



Social

- Preserves indigenous crafts & heritage
- Brings tier-2/3 & rural areas into tourism
- Multilingual access for all



Economic

- Credits recirculate in the local economy
- Hotels keep 100% of direct-booking margin
- Crowd balancing paid by hotels, not subsidies



Environmental

- Less strain on fragile spots (Ladakh, Shimla, Munnar)
- Rewards off-peak & eco-friendly transit
- Supports SDG 8 & SDG 12



Safety & privacy

- Offline SOS in no-network areas
- DPDP Act 2023: no PII, anonymised H3 counts
- Weather-aware re-planning

*Target outcomes to be validated in the pilot; microservice-based dynamic pricing has shown +22% revenue in published research (Barua & Kaiser, 2024).

Details / Links of the reference and research work



Data, platforms & policy

- Ministry of Tourism data portal (tourist footfall) data.tourism.gov.in
- OpenStreetMap (heritage sites, transit nodes) openstreetmap.org
- OpenWeatherMap API (weather-aware plans) openweathermap.org
- Antonio et al. (2019) Hotel booking demand datasets, Data in Brief 22 [doi:10.1016/j.dib.2018.11.126](https://doi.org/10.1016/j.dib.2018.11.126)
- Baileys WhatsApp Web library (our gateway) github.com/WhiskeySockets/Baileys
- Digital Personal Data Protection Act 2023 meity.gov.in
- UN Sustainable Development Goals 8 & 12 sdgs.un.org/goals



Research papers

- Barua & Kaiser (2024) Microservices dynamic pricing in travel: +22% revenue, 17% faster pricing [arXiv:2411.01636](https://arxiv.org/abs/2411.01636)
- Gössling & Mei (2025) AI and sustainable tourism: risks & opportunities for the SDGs, Current Issues in Tourism [doi:10.1080/13683500.2025.2477142](https://doi.org/10.1080/13683500.2025.2477142)
- Vickrey (1961) Counterspeculation, auctions & sealed tenders, J. of Finance (Demand Exchange) [doi:10.1111/j.1540-6261.1961.tb02789.x](https://doi.org/10.1111/j.1540-6261.1961.tb02789.x)
- Cifuentes (1992) Tourism carrying-capacity method, CATIE (Capacity Guard)
- Chen & Guestrin (2016) XGBoost: a scalable tree boosting system, KDD [doi:10.1145/2939672.2939785](https://doi.org/10.1145/2939672.2939785)
- Taylor & Letham (2018) Forecasting at scale (Prophet), The American Statistician [doi:10.1080/00031305.2017.1380080](https://doi.org/10.1080/00031305.2017.1380080)

Our prototype base: working multi-tenant WhatsApp AI gateway (Node.js + Baileys) with per-tenant AI prompts; codebase security review done and remediation plan defined (see Feasibility).

Competitors studied: OTAs (MakeMyTrip, Booking.com: 15–30% commission) · AI planners (Mindtrip.ai: no hotel link) · state e-pass systems (cap entry, don't redirect). **None of these offers a hotel-bid Demand Exchange or a per-hotel WhatsApp AI desk.**